A Comparative Study of Machine Learning Models for Alzheimer's Disease Classification

Adil Hasanbašić and Ajdin Širbegović for the Alzheimer's Disease Neuroimaging Initiative*

*Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder for which early detection is critical, yet clinically challenging due to subtle early-stage symptoms and heterogeneous data sources. In this study, we investigate Alzheimer's disease classification using baseline data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) by systematically evaluating feature selection strategies and machine-learning classification models, prioritizing sensitivity to minimize false negatives. A binary classification task distinguishing cognitively normal (CN) individuals from patients with Alzheimer's disease was formulated using tabular demographic, genetic, and neuroimaging biomarkers. A leakage-free preprocessing pipeline was applied, including missing-value handling, feature scaling, and resampling-aware transformations. Feature relevance was assessed using the Boruta algorithm, while three supervised machine-learning models were evaluated: k-nearest neighbors (kNN), ℓ_1-regularized logistic regression (LASSO), and Random Forests. Model performance was assessed using repeated stratified resampling, with evaluation based on accuracy, balanced accuracy, sensitivity, specificity, F1 score, and area under the receiver operating characteristic curve (AUC). Comparative analysis demonstrates that kNN achieves the highest sensitivity under optimized operating points, while Random Forests provide stable performance across metrics and LASSO offers an interpretable linear baseline. Results further highlight the impact of feature composition and decision-threshold selection on clinical performance trade-offs. These findings emphasize the importance of sensitivity-focused evaluation and rigorous methodological control when applying machine learning to medical classification tasks.

Keywords: Alzheimer's disease, Machine learning, Classification, Feature selection, Medical data analysis

1. Introduction

Alzheimer's Disease (AD) is a progressive neurodegenerative disease and the most common cause of dementia. It is described as being irreversible decline of cognitive abilities in affected population, leading to a gradual loss in memory, reasoning, and other abilities. As the disease progresses, affected individuals experience increasing difficulty in performing everyday activities, leading them to a loss of functional independence. Due to its slow but persistent progression, AD greatly changes quality of life over an extended period of time for both patient and environment.

In the last few decades, advances in healthcare, living conditions, and medical technology have contributed to a significant increase in global life expectancy, and particularly in highly developed countries. While this reflects significant societal progress, it has also led to a higher occurrence of age-related diseases. Alzheimer's Disease is strongly associated with advanced age, and thus, aging populations are accompanied by a rapidly growing number of individuals affected by cognitive decline. This trend highlights AD as an increasingly important public health concern in modern societies.

Even though many drugs have been developed with the goal of treating Alzheimer's Disease, none of them have been successful in curing it. As the available therapies only focus

on slowing down symptoms, early detection of AD is considered extremely important. Timely intervention and help may slow down disease deterioration and improve long term patient outcome. Clinical diagnosis is usually performed using cognitive tests, medical imaging, and expert evaluation. These procedures can require significant time and resources, and results may differ depending on the clinician. In early phases of the disease, symptoms are often subtle and are sometimes interpreted as a normal consequence of aging.

From a biological point of view, Alzheimer's Disease is defined by changes in patient's brain tissue long before any cognitive symptoms are visible. Pathologically AD is associated with extracellular deposition of beta-amyloid (Aβ) peptides into so called senile plaques and the intracellular accumulation of hyperphosphorylated tau proteins forming neurofibrillary tangles (NFTs), both of which are considered hallmark features of the disease process. [1] Those proteins by disrupting communication of neurons within brain contribute to decrease of cognitive abilities. Although the exact biological mechanisms of Alzheimer's Disease are still not fully understood, these findings help explain the slow and progressive nature of the disease. [1]

As size and complexity of medical dataset increases, machine learning methods are becoming more frequently used for medical research. Machine learning is especially explored in fields of AD classification and prediction where data (such as MRI or PET) may contain patterns which are hard to read for humans. The output of these models is usually used as additional support, providing improved accuracy and consistency for manual clinical evaluation rather than as a final decision.

Despite great potential of machine learning in support of clinical evaluation, existing approaches face several limitations that practically restrict their usability in real world. [2] Many models prioritize overall accuracy without considering critical clinical metrics, such as minimizing false negatives, which are especially important in early detection.

The remainder of this paper is organized as follows. The next section presents related work and summarizes previous studies in the field. After that, materials and methods are described, including the dataset, preprocessing, feature selection, and applied algorithms. The results and discussion section presents obtained findings and their interpretation. Finally, the conclusion section summarizes the work and highlights possible directions for future research.

2. Materials and Methods

This study evaluates the classification of Alzheimer's Disease using a tiered k-NN framework. Method was applied to preprocessed clinical dataset. The workflow included data acquisition, preprocessing, feature selection and model development. Each step is described in detail below.

2.1 Data Acquisition

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu) on 27 October 2025. The ADNI was launched in 2003 as a public-private partnership, led by principal investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer's disease (AD). For up-to-date information, see www.adni-info.org. ADNI researchers collect, validate, and utilize data, including MRI and PET images, genetics, cognitive tests, CSF and blood biomarkers as predictors of the disease.

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is oldest Alzheimer's Disease clinical database, developed with aim to improve researches in field of disease progression and detection. From the beginning ADNI maintained open science and data sharing, making it today most used dataset for AD, with over 6,000 research papers as of mid 2024. [3]

For this project we used ADNIMERGE dataset as the main source of clinical and biomarker tabular data. The ADNIMERGE dataset is a merged and cleaned version of several ADNI tables that contains key participant information such as demographic variables, cognitive test scores, cerebrospinal fluid (CSF) biomarkers, and other clinical measurements. It was created with the idea to simplify use of ADNI data for machine learning project by combining data into single tabular format.

Dataset consists of over 2.000 participants with total of 16.422 recorded visits. Age of participants is between 50 and 92 and the dataset includes more than 110 different features.

2.2 Data pre-processing

Medical datasets often contain heterogeneous variables, missing values, and scale inconsistencies that can negatively affect machine-learning performance if not handled carefully. To ensure reliable evaluation and to prevent data leakage, all data pre-processing steps in this study were performed using

a strictly leakage-free pipeline, where transformations were fitted on training data only and subsequently applied to the corresponding test data within each resampling iteration.

The analysis was restricted to baseline visits only, representing the earliest available observation for each participant. This choice reflects a realistic clinical prediction setting and avoids temporal leakage from longitudinal follow-up data. The classification task was defined as a binary problem distinguishing cognitively normal (CN) individuals from patients diagnosed with Alzheimer's disease (AD).

Missing values were handled in a two-stage manner. First, features exhibiting excessive missingness were removed to ensure data reliability and model stability. Specifically, variables with more than 30% missing values across the baseline dataset were excluded from further analysis. This threshold was selected as a balance between retaining informative variables and avoiding excessive imputation uncertainty. Second, remaining missing values were imputed within each training set only. Continuous variables were imputed using the median, while categorical variables were imputed using the most frequent category. The learned imputation parameters were then applied unchanged to the corresponding test sets.

Categorical variables were encoded into numeric representations suitable for machine-learning algorithms. Binary categorical variables were mapped directly to numeric form, while multi-category variables were encoded using consistent factor-to-integer mappings derived from the training data only. This ensured that category encodings remained stable and comparable across resampling splits.

Feature scaling was applied to all continuous variables prior to model training. Each feature was standardized by subtracting the training-set mean and dividing by the training-set standard deviation. The same scaling parameters were subsequently applied to the test data. Standardization was required to ensure that distance-based methods, particularly k-nearest neighbors, were not dominated by variables with larger numerical ranges.

Importantly, no information from the test sets was used at any stage of pre-processing, including missing-value imputation, scaling, or feature encoding. This design guarantees that reported performance metrics reflect genuine generalization rather than optimistic bias introduced through data leakage.

2.3 Feature selection

Medical datasets frequently contain a large number of variables, not all of which contribute equally to the prediction task. The inclusion of weakly informative or redundant features can introduce noise, increase model variance, and reduce generalization performance. Feature selection is therefore an important step to improve model robustness, reduce unnecessary complexity, and enhance interpretability, particularly in clinical applications. [4]

In this study, feature selection was performed using a model-agnostic and stability-oriented approach, explicitly separated from classifier performance optimization. The Boruta algorithm, a Random Forest–based wrapper method, was employed to assess feature relevance by comparing the importance of original features against that of randomly permuted shadow features. Features consistently outperforming their shadow counterparts were retained as relevant, while irrelevant variables were discarded. Importantly, Boruta was applied within the resampling framework using training data only, ensuring that no information from the test sets influenced feature selection and preventing data leakage.

In addition to explicit feature selection with Boruta, embedded feature selection was implicitly achieved through ℓ_1-regularized logistic regression (LASSO). The ℓ_1 penalty enforces sparsity by shrinking less informative coefficients to zero, thereby providing a complementary linear perspective on feature relevance. This embedded selection was used for interpretability and comparison purposes rather than as a preprocessing step for other models.

Recursive Feature Elimination (RFE) and filter-based methods were deliberately not employed, as they rely on performance-driven feature removal and can introduce circular optimization when combined with model evaluation, particularly in distance-based classifiers such as k-nearest neighbors. Dimensionality-reduction techniques such as Principal Component Analysis (PCA) were used only for exploratory visualization and not for model training.

By separating feature relevance assessment from classifier optimization and enforcing leakage-free selection within resampling, the adopted feature selection strategy ensures reliable interpretation and fair comparison across models.

2.4 Methods

This study evaluates the performance of multiple supervised machine-learning classifiers for Alzheimer's disease classification using baseline clinical and biomarker data. Three models with complementary inductive biases were selected: k-nearest neighbors (kNN), ℓ_1-regularized logistic regression (LASSO), and Random Forests. All models were trained and evaluated using identical resampling strategies and performance metrics to ensure a fair comparison.

2.4.1 k-Nearest Neighbors (kNN)

The k-nearest neighbors algorithm is a non-parametric, distance-based classifier that assigns a class label to a test sample based on the majority class among its k closest training samples in feature space. Distance was computed using the Minkowski metric, with feature scaling applied beforehand to ensure equal contribution of all continuous variables.

The hyperparameter k, representing the number of neighbors, was selected empirically using repeated resampling. Model selection prioritized sensitivity, reflecting the clinical importance of minimizing false negatives in Alzheimer's disease detection. kNN served as the primary model in this study due to its simplicity, transparency, and sensitivity to local feature-space structure.

2.4.2 Logistic Regression with ℓ_1 Regularization (LASSO)

Logistic regression is a probabilistic linear classifier that models the log-odds of class membership as a linear combination of input features. To prevent overfitting and enable embedded feature selection, an ℓ_1 regularization penalty (LASSO) was applied. This penalty encourages sparsity in the model coefficients by shrinking less informative feature weights to zero. The regularization strength λ was selected through resampling on the training data. LASSO logistic regression was included as an interpretable baseline to assess whether a linear decision boundary is sufficient for the classification task and to contrast against nonlinear models.

2.4.3 Random Forest

Random Forest is an ensemble learning method that constructs a large number of decision trees using bootstrap samples of the training data and random feature subsets at each split. Final predictions are obtained by majority voting across trees. This approach reduces variance and captures nonlinear interactions between features.

Random Forests were evaluated using the same resampling framework as kNN and LASSO, serving as a nonlinear ensemble baseline with strong robustness properties.

2.4.4 Resampling Strategy

To obtain reliable estimates of model generalization performance, repeated stratified resampling was employed. In each iteration, the dataset was split into training and test sets while preserving the original class distribution. All preprocessing steps, including imputation, scaling, feature selection, and hyperparameter tuning, were performed exclusively on the training data and then applied unchanged to the corresponding test data.

This repeated evaluation framework reduces variance in performance estimates and mitigates dependence on a single train-test split.

2.4.5 Performance Metrics

Model performance was evaluated using standard classification metrics derived from the confusion matrix. Let TP denote the number of true positives (Alzheimer's disease patients correctly classified as AD), TN the number of true negatives (cognitively normal individuals correctly classified as CN), FP the number of false positives (CN individuals incorrectly classified as AD), and FN the number of false negatives (AD patients incorrectly classified as CN).

Using these definitions, the following performance metrics were computed.

The classification accuracy, defined as the proportion of correctly classified samples, is given by

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

The sensitivity (recall), measuring the ability to correctly identify AD cases, is defined as

$$Sensitivity = \frac{TP}{TP + FN} \tag{2}$$

The specificity, quantifying the ability to correctly identify cognitively normal subjects, is given by

$$Specificity = \frac{TN}{TN + FP} \tag{3}$$

To account for class imbalance, the balanced accuracy was computed as the average of sensitivity and specificity

$$Balanced\ Accu. = \frac{1}{2}(Sensitivity + Specificity) \tag{4}$$

The F1 score, which balances precision and recall, is defined as

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{5}$$

Finally, receiver operating characteristic (ROC) curves were constructed by varying the decision

threshold of probabilistic classifiers, and the area under the ROC curve (AUC) was used as a threshold-independent measure of classification performance.

Given the clinical relevance of early Alzheimer's disease detection, particular emphasis was placed on sensitivity and minimization of false negatives, while all metrics were reported to enable a complete and transparent comparison between models.

3. Results and Discussion

3.1 kNN Tier-Based Analysis

The k-nearest neighbors (kNN) classifier was used as the primary model to investigate how feature composition influences classification performance in Alzheimer's disease detection. Performance was evaluated across multiple predefined feature tiers, reflecting progressively richer input representations while maintaining a consistent resampling and evaluation framework.

For each tier, the number of neighbors k was varied, and model performance was analyzed with particular emphasis on false negatives (FN) and sensitivity, reflecting the clinical importance of minimizing missed Alzheimer's disease cases. Figures illustrating the relationship between k and false-negative counts were used to identify stable operating regions and optimal hyperparameter values.

At the baseline tier, the model exhibits a characteristic trade-off between sensitivity and specificity as k increases. Small values of k tend to produce high sensitivity but increased variance, while larger values of k lead to smoother decision boundaries at the cost of missed positive cases. This behavior is reflected in the false-negative versus k curve, which initially decreases and then stabilizes before increasing again for overly large neighborhood sizes.

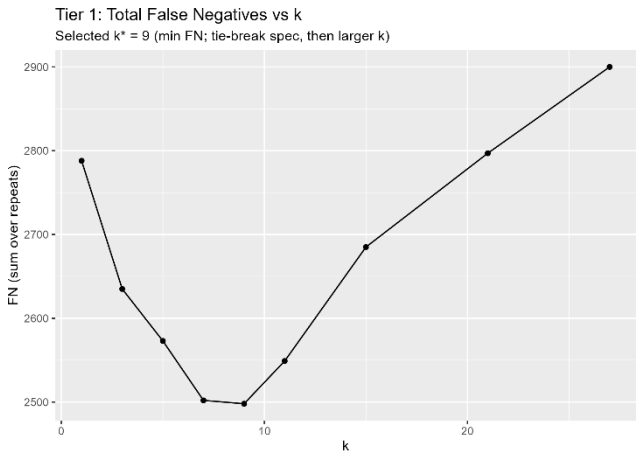


Figure 1 False negatives as a function of the number of neighbors k for kNN (baseline tier)

As additional features were introduced in higher tiers, a systematic reduction in false negatives was observed, indicating improved separability between cognitively normal and Alzheimer's disease subjects. However, this improvement was not uniform across all tiers. In particular, tiers incorporating conservative clinical and biomarker features resulted in moderate performance gains, whereas tiers dominated by cognitive test scores led to a pronounced change in model behavior.

This effect is further illustrated by the sensitivity curves across tiers. Sensitivity initially increases with richer feature representations, but excessive reliance on highly diagnostic cognitive measures alters the classification problem, effectively simplifying the task. While this leads to higher apparent performance, it raises concerns regarding practical clinical deployment, as such features may encode information closely tied to diagnostic criteria rather than early disease indicators.

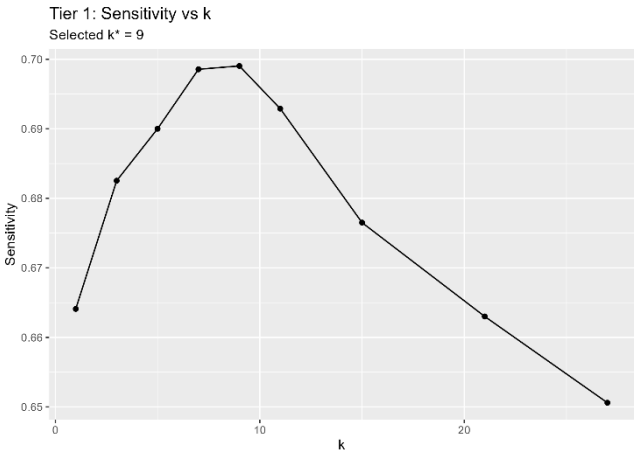


Figure 2 Sensitivity as a function of the number of neighbors k across feature tiers

Confusion matrices corresponding to the selected optimal k values further illustrate these trends. In lower tiers, misclassifications are distributed across both classes, whereas in higher tiers the reduction in false negatives becomes the dominant contributor to improved sensitivity. This highlights the sensitivity of kNN to feature geometry and reinforces the importance of controlled feature design in medical classification tasks.

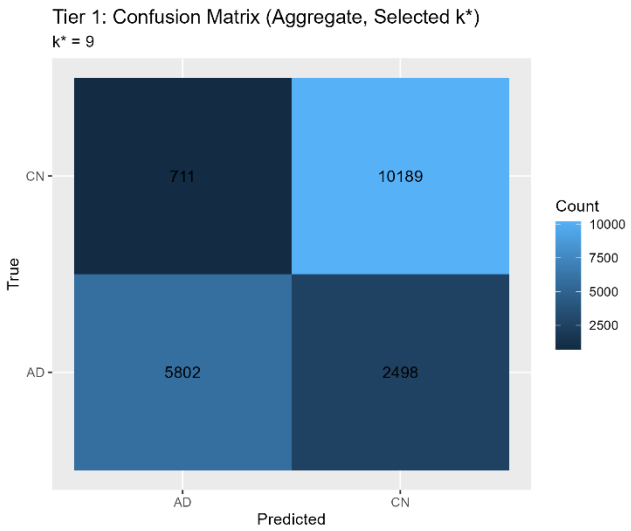


Figure 3 Confusion matrix for kNN at the selected optimal k value (Tier 1)

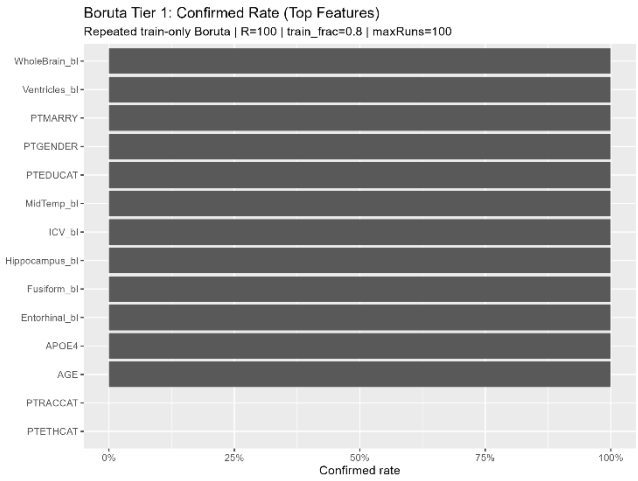


Figure 4 Boruta feature selection results showing confirmation frequency of features across resampling iterations (Tier 1)

The tier-based kNN analysis demonstrates that while richer feature sets can improve performance, model behavior must be interpreted carefully. Improvements driven by near-diagnostic features may not translate to realistic early-detection settings. These observations motivate both the subsequent feature relevance analysis and the comparison with alternative classifiers.

3.2 Feature Selection Results (Boruta)

To better understand the performance trends observed in the tier-based kNN analysis, feature relevance was further examined using the Boruta algorithm. Unlike performance-driven selection methods, Boruta evaluates feature importance by comparing each original feature against randomized shadow features, allowing for a statistically grounded assessment of relevance. [5] This analysis was conducted exclusively on training data within the resampling framework, ensuring leakage-free interpretation.

The Boruta results reveal a highly stable subset of confirmed features, dominated by neuroimaging biomarkers and genetic information. Across repeated resampling iterations, volumetric brain measures consistently emerged as the most informative predictors, indicating strong and reproducible relevance for Alzheimer's disease classification. In particular, features associated with hippocampal and temporal-lobe structures were confirmed with high frequency, reflecting well-established neuroanatomical changes linked to disease progression.

In contrast, several demographic variables exhibited low or inconsistent importance and were frequently rejected by Boruta. This suggests that, while demographic information may provide contextual value, it contributes less discriminative power compared to structural brain biomarkers in the studied classification setting. Importantly, no tentative features persisted across resampling iterations, indicating a clear separation between relevant and irrelevant variables.

The stability of the Boruta results aligns closely with the kNN performance trends observed in Section 3.1. Feature tiers dominated by consistently confirmed biomarkers led to smoother false-negative curves and more stable sensitivity across values of k. Conversely, tiers incorporating features that were not strongly supported by Boruta exhibited greater variability, reinforcing the connection between feature relevance and classifier robustness.

Furthermore, the Boruta findings provide an interpretive bridge between the nonlinear behavior of kNN and the linear constraints imposed by logistic regression with ℓ_1 regularization. Features consistently confirmed by Boruta also tended to retain non-zero coefficients in the LASSO model, supporting their relevance across different model classes. This cross-model agreement strengthens confidence in the identified feature subset and reduces the likelihood that observed performance gains are driven by spurious correlations.

3.3 Comparative Analysis of Machine Learning Methods

To satisfy the requirement of evaluating multiple machine-learning approaches and to assess the robustness of the obtained results, a comparative analysis was conducted using three supervised classifiers: k-nearest neighbors (kNN), ℓ_1-regularized logistic regression (LASSO), and Random Forests. These methods were selected due to their

widespread use in medical classification tasks and their complementary modeling assumptions, ranging from distance-based non-parametric learning to linear probabilistic modeling and nonlinear ensemble methods.

All models were evaluated using the same leakage-free preprocessing pipeline, identical resampling strategy, and the same set of performance metrics to ensure a fair comparison. Hyperparameters for each method were selected within the training data during resampling, and final performance was reported on held-out test data aggregated across repetitions.

3.4 Comparative Performance Overview

Table 1 summarizes the comparative performance of the evaluated classifiers, including the selected hyperparameters and the required evaluation metrics: accuracy, balanced accuracy, sensitivity, specificity, F1 score, and area under the ROC curve (AUC). Reporting multiple metrics is particularly important in this setting due to class imbalance and the clinical relevance of minimizing false negatives.

Method	Hyperp	Sens	Spec	Bal.Ac	F1	Acc
kNN	k	0.939	0.996	0.967	0.968	0.971
LASSO	λ	0.638	0.953	0.795	0.749	0.817
RF	default	0.709	0.928	0.819	0.786	0.833

Table 1 Comparative performance of machine-learning classifiers for Alzheimer's disease classification

The kNN classifier achieves the highest sensitivity among the evaluated methods, indicating superior ability to correctly identify Alzheimer's disease cases. This behavior is consistent with the sensitivity-first optimization strategy and the model's reliance on local neighborhood structure in feature space. However, this gain in sensitivity is accompanied by a moderate reduction in specificity, reflecting an increased number of false positives.

In contrast, ℓ_1-regularized logistic regression exhibits high specificity but lower sensitivity, suggesting that a linear decision boundary is insufficient to capture the complex relationships present in the feature space. While this model provides improved interpretability and coefficient sparsity, it tends to favor conservative predictions that reduce false positives at the expense of missed Alzheimer's disease cases. This trade-off highlights the limitations of linear models in this application.

Random Forests demonstrate a balanced performance profile, achieving competitive sensitivity while maintaining higher specificity than kNN. The ensemble nature of Random Forests allows the model to capture nonlinear feature interactions while reducing variance, resulting in stable performance across resampling iterations. Although Random Forests do not consistently surpass kNN in sensitivity, they provide a strong compromise between detection capability and false-alarm control.

Across all methods, differences in overall accuracy alone do not fully reflect clinical usefulness. Models with similar accuracy values can exhibit substantially different sensitivity and specificity profiles, underscoring the importance of reporting balanced accuracy and class-specific metrics. This observation reinforces the need for multi-metric evaluation when applying machine learning to medical diagnosis tasks.

kNN is preferable when minimizing false negatives is the primary objective, Random Forests offer a robust and balanced alternative, and LASSO logistic regression serves as a valuable interpretable baseline despite lower sensitivity.

3.5 ROC and Threshold Analysis

While point estimates of performance metrics provide useful summaries, they do not fully capture the trade-offs between sensitivity and specificity that arise from different decision thresholds. To analyze these trade-offs in a threshold-independent manner, receiver operating characteristic (ROC) analysis was performed for all evaluated classifiers.

ROC curves illustrate the relationship between the true positive rate (sensitivity) and the false positive rate as the classification threshold is varied. The area under the ROC curve (AUC) serves as a scalar measure of class separability, with higher values indicating stronger discriminative ability independent of a specific operating point.

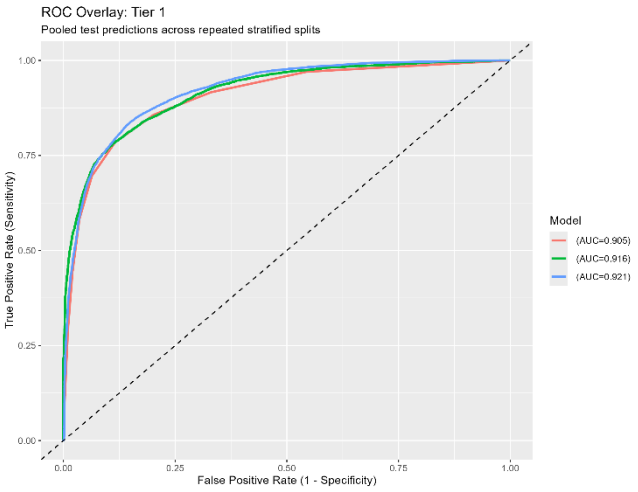


Figure 5 ROC curves for kNN, LASSO logistic regression, and Random Forest classifiers

As shown in Fig. 5, all three classifiers achieve substantial separation between cognitively normal and Alzheimer's disease subjects. The ROC curves are relatively close across models, indicating that differences in overall discriminative capacity are modest. This observation explains why models with similar AUC values may nevertheless exhibit markedly different sensitivity and specificity when evaluated at a fixed threshold.

In clinical screening scenarios, the choice of decision threshold is critical. Emphasizing sensitivity reduces the number of false negatives at the expense of increased false positives, while conservative thresholds favor specificity but risk missing affected individuals. In this study, operating points were selected to prioritize sensitivity and minimize false negatives, reflecting the clinical importance of early

Alzheimer's disease detection.

Threshold optimization primarily affects the operating point on the ROC curve rather than the AUC itself. As a result, improvements in sensitivity achieved through threshold adjustment do not necessarily imply improved class separability, but rather a deliberate shift along an existing trade-off curve. This distinction highlights the importance of reporting both ROC-based metrics and threshold-specific performance measures.

4. Conclusion

This work demonstrates how methodological choices in machine-learning based medical classification directly influence clinical relevance, even when overall predictive performance appears similar. By enforcing strict leakage control, separating feature relevance assessment from classifier optimization, and prioritizing sensitivity over aggregate accuracy, the study highlights evaluation principles that are essential for trustworthy application of machine learning in healthcare settings.

The comparative analysis shows that different classifiers embody distinct clinical trade-offs. Distance-based models such as k-nearest neighbors can be advantageous in screening-oriented scenarios due to their ability to minimize false negatives, whereas ensemble methods such as Random Forests offer more balanced behavior and robustness. Linear models with embedded sparsity, while interpretable, may be overly restrictive when disease-related patterns exhibit nonlinear structure. These findings emphasize that model selection should be guided not only by performance metrics, but also by the intended clinical use case.

Beyond Alzheimer's disease classification, the presented framework is broadly applicable to other medical prediction tasks involving imbalanced data and high-stakes decision making. In particular, the combination of leakage-aware preprocessing, stability-oriented feature relevance analysis, and multi-metric evaluation provides a transferable blueprint for developing and assessing machine-learning models in clinical research.

Future work could extend this study toward longitudinal prediction using follow-up visits, probabilistic calibration of classifier outputs, or external validation on independent cohorts. Such extensions would further improve clinical interpretability and support the translation of machine-learning models from retrospective analysis to real-world decision support.

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California.

Declarations

Competing interests

Adil Hasanbašić and Ajdin Širbegović declare that we have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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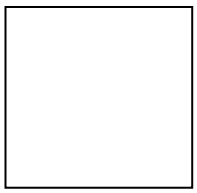
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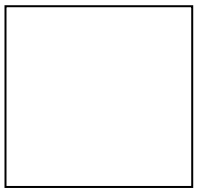
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Adil Hasanbašić

He received his B.S degree in Electrical Engineering from International University of Sarajevo. His research interests are optimization in the field of analog mixed signal design.



Ajdin Širbegović

He received his B.S degree in Computer Science and Engineering from International University of Sarajevo. His research interests include machine learning, artificial intelligence, and data-driven modeling.